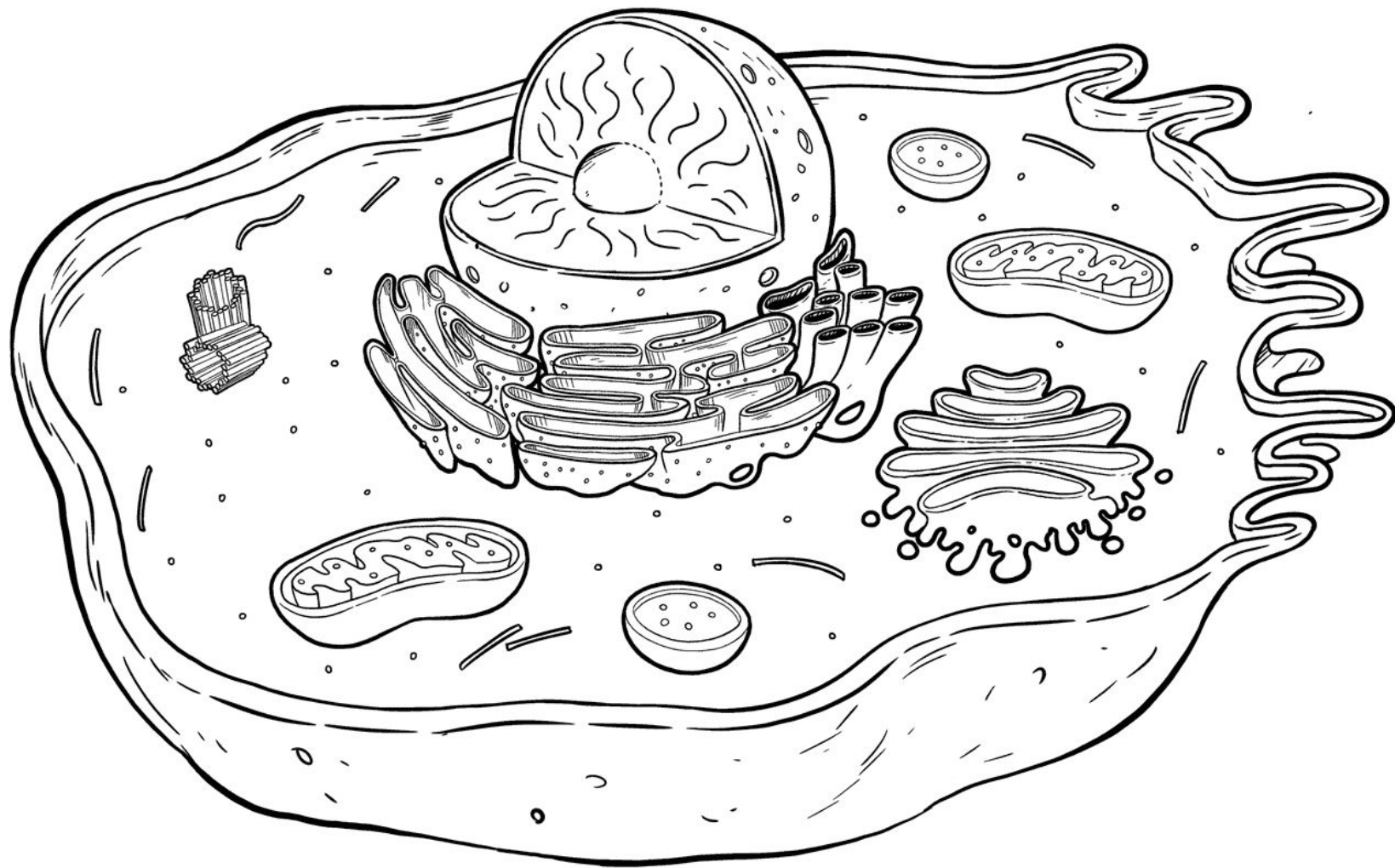




The Cell Organelles and their Functions

Nucleus – The cell nucleus is an organelle that contains most of the cell's genetic material. The nucleus regulates cell growth and metabolism.

Nucleolus – The nucleolus is made of RNA and proteins. It transcribes and modifies RNA.

Lysosomes – The lysosomes contain enzymes that break down biomolecules. They act as the cell's waste disposal.

Chromatin – Chromatin are macromolecules made up of DNA, protein and RNA.

Centrioles – The centrioles help with cytokinesis. (The splitting of a cell)

Cytoskeleton – The cytoskeleton gives a cell its shape, and prevents it from deforming.

Cytoplasm – The cytoplasm is made up of cytosol, which is a gel-like substance. It is 80% water.

Plasma Membrane – Also known as the cell membrane, the plasma membrane is a selectively permeable wall that separates the cell interior from the outside environment.

Ribosomes – The ribosomes are made of protein and RNA. They convert genetic material into protein.

Rough Endoplasmic Reticulum – The rough endoplasmic reticulum manufactures enzymes and proteins.

Smooth Endoplasmic Reticulum – The smooth endoplasmic reticulum manufactures lipids, phospholipids, and steroids.

Golgi Apparatus – The golgi apparatus, also known as the golgi bodies, packages protein inside a cell, in preparation for secretion.

Mitochondria – The mitochondria are a cell's power plants. They generate ATP.